



NATIONAL OCCUPATIONAL STANDARD FOR AN ANIMAL BREEDER- GENETICST

**NOS.AB.01
FIRST EDITION**

APPROVING AUTHORITY

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ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards shall be revised after every **5 years**, or whenever necessary, by the issue of either amendments or of revised editions. It is important that users of National Occupational Standards (NOS) ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Agriculture National Occupational Standards Development Team, upon which the following organisations were represented:

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2. Aquaculture Development Association of Zambia
3. CropLife Zambia
4. Ministry of Fisheries and Livestock (Department of Fisheries)
5. Golden Valley Agricultural Research Trust (GART)
6. Katete College of Agricultural Marketing
7. Ministry of Agriculture (Department of Agriculture)
8. Mulungushi University (MU)
9. Natural Resources Development College (NRDC)
10. University of Zambia (UNZA)
11. Veterinary Council of Zambia
12. Zambia Agriculture Research Institute (ZARI)
13. Zambia National Farmers Union (ZNFU)
14. Zambia Seed Trade Association (ZASTA)
15. Zambia Qualifications Authority

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TABLE OF CONTENTS

FOREWORD	iv
JUSTIFICATION	iv
ACRONYMS AND ABBREVIATIONS	v
GLOSSARY OF TERMS	vi
1. OVERVIEW	1
2. SCOPE.....	2
3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)	2
4. UNITS AND ELEMENTS.....	2
5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS.....	17
6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER.....	17
7. WORKING CONDITIONS/ENVIRONMENT	17
8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE	18
9. PHYSICAL DEMANDS ON THE BODY	18
ANNEX A.....	19
ANNEX B.....	20

FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to “***provide for the development and implementation of a national qualifications framework; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing***”.

Among other functions, ZAQA is responsible for “***determining national standards for any occupation***”, through various sector specific National Occupational Standards Development Teams (NOSDTs) of experts composed of representation from appropriate authorities, government departments, industry, academia, regulators, consumer associations and non-governmental organizations, etc.

This National Occupational Standard (NOS) has been developed by the Agriculture National Occupational Standards Development Team in accordance with the procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula and learning programmes, in various sectors where the occupation exists.

JUSTIFICATION

Animal breeding ensures a continuous improvement of farm animals, generation after generation. Desired animal traits are measured and improved, and the best animals are used as parent stock. An Animal Breeder-Geneticist provides researchers and livestock farmers with a next generation of animals, making the job an important one in animal breeding services.

This occupation is vital for conservation and management of superior animal genetic traits or resources. The genetic diversity of our livestock species is in a state of continual decline, and the animal genetic resources that remain are often not used as efficiently as they could be, in addressing nutritional deficiencies and eradicating poverty. Therefore, an Animal Breeder will provide technical guidance in establishment of breeding programs that will conserve our livestock genetic resources and stimulate optimal usage of these resources.

ACRONYMS AND ABBREVIATIONS

AB	Animal Breeder - Geneticist
AI	Artificial Insemination
AIZ	Agricultural Institute of Zambia
CS	Core Skill
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organizational Knowledge
PC	Performance Criteria
PS	Professional Skill
RK	Regulatory Knowledge
RPL	Recognition of Prior Learning
TK	Technical Knowledge
VCZ	Veterinary Council of Zambia
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Function: is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles, which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

1. OVERVIEW

This section provides a brief summary of the Animal Breeder – Geneticist job profile, taking into account what is required of the occupation from an academic and professional training perspective, as well as target sectors

NOS Code	NOS. AB.01
Occupation	Animal Breeder – Geneticist
Job Title	Animal Breeder – Geneticist
Job Description	An Animal Breeder-Geneticist works with research management and technical production teams to evaluate, select and mate animal populations in order to maximize desirable genetic traits and consequently improve/optimize production and productivity.
Job Purpose	The job role is responsible for ensuring proper livestock breeding management through evaluating the genetic value of animals (at phenotypic or molecular levels), deciding which animals are chosen as parents and the extent to which they are used (Selection), and deciding which males are mated to which females (Mating Systems) in order to improve populations of animals, thereby improving animal value and well-being.
ZQF Level	9
Sector	Agriculture
Sub sector	Livestock (Animal Breeding)
Other Economic Sector(s) in which the Occupation is Practiced	Education Sector (Academia, Research Industry), Manufacturing Sector (Food industry), Agricultural Sector (Extension Education Services, Breeding Farms)
Other Similar Jobs that can be performed by an Animal Breeder – Geneticist	Agricultural Sector (Zoologist, Biologist, Animal Biotechnologist, Animal Welfare Specialist, Laboratory Scientist)
Minimum Educational Job Entry Qualification(s)	Master's Degree in Animal breeding, or its equivalent
Practicing License Requirements (if any)	VCZ / AIZ
Training/RPL	Bachelor's Degree in Agricultural Sciences and/or Biological Sciences, or equivalent
Minimum Job Entry Age	24 years
Prior Experience	None required
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This National Occupational Standard specifies the fundamental knowledge and understanding, skills and competences that an Animal Breeder - Geneticist must possess to be successful in his/her job role. It is applicable to senior management working on Animal Breeding programs in public or private organizations, non-governmental organizations or self-employed businesses.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess:

- Dependability
- Creativity
- Integrity and self-control
- Complex problem-solving skills
- Analytical and scientific software skills
- Mathematics skills
- Critical Thinking skills
- Active learning and listening skills
- Judgement and decision-making skills
- Record-keeping skills
- Integrity and respect for confidentiality
- Interpersonal skills
- Commercial Awareness
- Meticulous attention to details
- Ability to communicate effectively and clearly
- Self-motivated and team worker
- Ability to plan and prioritize,
- Quality consciousness
- Determination and decisiveness
- Patience
- Flexibility
- Compassion for animals

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into four (04) Units representing the tasks that a job holder should undertake in his/her day to day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This unit is about research and management of Animal Breeding programs - Breeding Evaluation Process]

Unit No.	01
Unit Title	Planning and Implementation of Pre-Breeding Activities (Breeding Evaluation Process)
Description	This unit is about the prior knowledge and competence skills required by an Animal Breeder to demonstrate and supervise breeding program evaluations.
Scope	This unit covers the following: • Establishment of breeding objectives • Evaluation of animal populations in accordance with various Husbandry Management practices • Determination of risks and challenges associated with Animal Breeding
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Establishment of breeding objectives	To be competent, the individual must be able to: PC1. Identify problems for research investigation/analysis PC2. Establish clear breeding objectives which are the specific goals for the breeding program (SMART goals). These should clearly align with a farmer/stakeholder needs analysis. PC3. Exemplify which traits are of economic importance for the breeding program (desirable traits) PC4. Develop an action plan for Animal Breeding procedures and mobilize human resources and equipment needed. PC5. Read, interpret and implement national and international regulations pertaining to Animal Breeding
Evaluation of animal populations in accordance with various Husbandry Management practices	To be competent, the individual must be able to: PC6. Establish populations to be evaluated PC7. Conduct analysis of the animal populations (species and breeds) by clearly defining and documenting distinct breed populations and describing their external and production characteristics PC8. Demonstrate accurate and timely record keeping of animal inventory, performance, behaviour and conditions PC9. Ensure all routine management practices are followed ranging from standard medicine and feed regimes, proper facilities, biosecurity procedures etc PC10. Delegate overall co-ordination of breeding activities among co-workers/subordinates in line with standard procedures PC11. Ensure proper chain of command and flow of information between management and technical staff PC12. Assess performance of co-workers (subordinate staff)

Determination of risks and challenges associated with Animal Breeding	To be competent, the individual must be able to: PC13. Determine associated risks of a breeding program and mitigation measures, such as unforeseen disease outbreaks, PC14. Identify effects of seasonality on performance, PC15. Identify breeds that are potentially at risk of being depleted. PC16. Analyze the genetic makeup of animals at a molecular level, in order to discover which genes cause them to behave in certain ways (deleterious or beneficial).
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Organization legislation, standards, policies and procedures relevant to Animal Welfare OK2. Organization Quality, health and safety policies OK3. Organization environmental policies and bio-security practices OK4. Organization regulations and global best practices OK5. Organization vision, mission and strategic plan with regard to Animal Breeding.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Animal (anatomy, physiology, genotypes) and their effects on physical performance (It's genotype and performance attributes) TK2. Physical Environment (external non-genetic factors (nutrition, facilities, management practices etc) that influence the performance) TK3. Business and management principles involved in strategic planning, resource allocation, human resources modelling, leadership techniques, production methods, and coordination of people and resources TK4. Social norms and perceptions TK5. Biosafety and health best practices
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Government regulatory agency requirements for Animal welfare, health & safety. RK2. National pedigree databases as registered by the Herd-book Society of Zambia or other recognized agencies
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills The individual on the job must be able to: CS1. Write in English and give simple concise instructions. CS2. Record data of all observed traits of interest CS3. Record all activities undertaken and challenges faced

	CS4. Write and manage a personal daily task list as well as for subordinates CS5. Be conversant with data entry software like MS Excel CS6. Write required reports in a timely manner (monthly, quarterly or annual)
	Reading Skills
	The individual on the job must be able to: CS7. Read and interpret internal/external documents. CS8. Read and understand manuals, health and safety instructions, memos, other company documents. CS9. Read from different sources- books, journal publications, research articles on Animal Breeding.
	Oral Communication (Listening and Speaking skills)
B. Professional Skills	The individual on the job must be able to : CS10. Express statements or information clearly so that others can hear and understand. CS11. Communicate clearly during stakeholder round table meetings CS12. Participate in and understand the main points of simple discussions regarding Animal Breeding. CS13. Respond appropriately to any queries regarding Animal Breeding. CS14. Communicate clearly to possible donors/funders and possibly apply for grants
	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan for procurement of livestock for breeding in a timely manner PS4. Plan to arrange for necessary materials to be in place PS5. Plan and organise work to meet deadlines. PS6. Plan to have up-to-date records at all times PS7. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS8. Follow code of conduct. PS9. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS10 Recognize problems and search for solutions. PS11. Choose best methods to complete assigned tasks. PS12. Approach relevant authority when required. PS13. Judiciously use common sense in day to day activities

	Analytical Thinking
	The individual on the job must be able to: PS14. Apply domain knowledge, observations and data to select course of action to perform tasks PS15. Monitor breeding conditions and breeding livestock and suggest improvements if any.
	Critical Thinking
	The individual on the job must be able to: PS16. Use common sense and make judgements on day to day basis PS17. Critically evaluate information obtained from stakeholders and co-workers to perform day to day activities. PS18. Use reasoning skills to identify and resolve basic problems

UNIT 2 [This unit is about Animal Breeding Selection and Mating Systems Process]

Unit No.	02
Unit Title	Implementing Breeding Selection and Mating System Procedures
Description	This unit is about affecting genetic change by choosing replacements with the best set of genes (higher breeding values) to form the next generation, and critically determining which selected males are mated to which selected females, in order to improve particular traits and thereby improve value and wellbeing of animals
Scope	This unit covers the following: • Establishment of type and method of Selection for traits of economic importance • Evaluation of modes of gene action and analysis of Heritability and Repeatability Relationships with Phenotypes • Estimation of Breeding Values • Determination of mating decisions, making use of breed complementarity and hybrid vigour. • Analysis of negative effects of breeding on populations • Conducting servicing of selected animals • Monitoring of progress and disseminating information to farmers/stakeholders and/or public domain
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Establish type and method of selection for traits of economic importance	To be competent, the individual must be able to: PC1. Ascertain animals of higher genetic and phenotypic merit, to be chosen as parents for the next generation PC2. Demonstrate the type of selection to be used; phenotypic and/or pedigree-based selection

	PC3. Determine the method of selection (single or multiple trait selection) and procedures followed, in alignment with traits of economic importance. PC4. Determine factors that effectively influence selection PC5. Document all selection procedures of animals, according to their genealogy, characteristics and pedigree PC6. Establish which males naturally mate which females, and why. PC7. Apply knowledge of Artificial Insemination techniques and equipment use PC8. Delegate and document observations of animals in heat to detect approach of oestrus PC9. Institute an apt record performance system. This entails keeping accurate records on heats, birth intervals, pedigree, growth, carcass attributes, milk yield, litter size, PC10. Evaluate for causal variants (disease, performance, colour) associated with a trait PC11. Establish traceability of parentage PC12. Supervise genetic testing procedures (elaborate procedures)
Evaluate modes of gene action for specific traits and analyse Heritability and Repeatability Relationships with Phenotypes	To be competent, the individual must be able to: PC13. Determine the mode of gene action expressed in the performance of animals PC14. Analyse Heritability and its relationship to observed phenotypes/performance PC15. Evaluate repeatability and demonstrate its importance in culling decisions.
Estimation of Breeding Values	To be competent, the individual must be able to: PC16. Conduct estimation of breeding values using Animal Breeding software packages like ASREML, MTDFREML or equivalent. PC17. Demonstrate how breeding values are used in selection index criteria.
Determination of mating decisions, making use of breed complementarity and hybrid vigour.	To be competent, the individual must be able to: PC18. Evaluate populations for interbreeding or between breed selection. PC19. Implement mating decisions based on breed complementarity and hybrid vigour concepts.
Analysis of negative effects of breeding on populations	To be competent, the individual must be able to: PC20. Analyse for genetic inbreeding depression and identify traits affected PC21. Compute and interpret inbreeding coefficients for animal populations PC22. Compute and interpret correlation coefficients of inbreeding and fertility traits.

Monitoring of Progress and disseminating information to farmers/stakeholders and/or public domain	To be competent, the individual must be able to: PC23. Demonstrate strict monitoring and evaluation procedures of breeding program and submission of monthly, quarterly and annual report progression PC24. Disseminate all progress information reports of breeding program to farmers/stakeholders and public domain through print and electronic media like social media, journal articles, radio programs, sector professional reports etc.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Organization legislation, standards, policies and procedures relevant to Animal Welfare OK2. Organization Quality, health and safety policies OK2. Organization environmental policies OK3. Organization regulations and global best practices OK4. Business and management principles involved in strategic planning, resource allocation, human resource modelling, leadership techniques, production methods and coordination of people/resources
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Animal (anatomy, physiology, genotypes) and their effects on physical performance (It's genotype and performance attributes) TK2. Physical Environment (external non-genetic factors (nutrition, facilities, management practices etc) that influence the performance) TK3. Business and management principles involved in strategic planning, resource allocation, human resources modelling, leadership techniques, production methods, and coordination of people and resources TK4. Animal husbandry equipment (E.g. Animal pregnancy testing kits; Dehorners; Ovulation prediction monitors; Semen collection kits, Artificial insemination chute, etc) TK5. Livestock identification equipment such as Branding equipment; Ear tagging equipment; Microchip scanners; Tattoo equipment TK6. Biosafety and health best practices TK7. Reproductive behaviour and characteristics of different animal species TK8. Animal Breeding software (ASREML, MTDFREML etc) TK9. Electronic mail software (Gmail, Yahoo etc) TK10. Spreadsheet software (MS Excel) TK11. Mathematics (Arithmetic, algebra, geometry, calculus, statistics)

C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Government regulatory agency requirements for Animal welfare, health & safety.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions. CS2. Record data of all observed traits of interest CS3. Record all activities undertaken and challenges faced
	Reading Skills
	The individual on the job must be able to: CS4. Read and interpret internal/external documents. CS5. Read and understand manuals, health and safety instructions, memos, other company documents. CS6. Read from different sources- books, journal publications, and research articles on Animal Breeding.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS7. Express statements or information clearly so that others can hear and understand. CS8. Participate in and understand the main points of simple discussions regarding Animal Breeding. CS9. Respond appropriately to any queries regarding Animal Breeding.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response. PS3. Consider relative costs and benefits of potential actions to choose the most appropriate one
	Plan and Organise
	The individual on the job must be able to: PS4. Plan for procurement of livestock for breeding in a timely manner PS5. Plan to arrange for necessary materials to be in place PS6. Plan and organise work to meet deadlines. PS7. Plan to have up-to-date records at all times PS8. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS9. Follow code of conduct. PS10. Manage relationships with customers with intent on satisfying its requirements for service delivery. PS11. Carry out customer needs assessment and evaluation of customer satisfaction

	Problem Solving and Decision Making
	The individual on the job must be able to: PS12. Recognize problems and search for solutions. PS13. Choose best methods to complete assigned tasks. PS14. Approach relevant authority when required. PS15. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS16. Apply domain knowledge, observations and data to select course of action to perform tasks PS17. Monitor breeding conditions and breeding livestock and suggest improvements if any.
	Critical Thinking
	The individual on the job must be able to: PS18. Use common sense and make judgements on day to day basis PS19. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS20. Use reasoning skills to identify and resolve basic problems

UNIT 3 [This unit is about research and management of Animal Breeding programs – Promotion of Animal Breeding technologies and technical knowhow to stakeholders]

Unit No.	03
Unit Title	Promotion of Animal Breeding Technology Awareness Programs
Description	This unit is about the promotion and training of relevant and up-to-date Animal Breeding Technologies.
Scope	This unit covers the following: • Demonstration of Animal Breeding technologies and practices • Evaluation of Animal Breeding technology utilization • Monitoring and evaluation of post-training progress of stakeholders • Conduct desk studies on pressing issues/challenges regarding Animal Breeding technologies.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Demonstration of Animal Breeding technologies and practices	To be competent, the individual must be able to: PC1. Demonstrate suitable technologies (practiced on a global front) required for stakeholder needs such as DNA fingerprinting, sanger sequencing, use of genetic markers, Linkage analyses, Marker assisted selection, genomic selection, etc

	PC2. Determine appropriate season/time for breeding animals using chosen technologies PC3. Analyse challenges and risks likely to arise and have mitigation measures in place
Evaluation of Animal Breeding technology utilization	To be competent, the individual must be able to: PC4. Engage stakeholder round table meetings to elicit information on technologies being utilized PC5. Develop training curriculum and demonstrate farmer/stakeholder trainings on relevant technologies/new technologies on the market using appropriate methods and planned schedules. PC6. Document all activities and trainings and report accordingly.
Monitoring and evaluation of post-training progress of stakeholders	To be competent, the individual must be able to: PC7. Develop a monitoring and evaluation program for all trainings offered to farmers/stakeholders
Conduct desk studies on pressing issues/challenges regarding Animal Breeding technologies	To be competent, the individual must be able to: PC8. Identify challenges and pressing issues arising from use of Animal Breeding technologies PC9. Conduct studies on causes of possible challenges faced with regards Animal Breeding technologies used. PC10. Report the findings via stakeholder meetings, print and electronic media publications.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Organization legislation, standards, policies and procedures relevant to Animal Welfare OK2. Organization Quality, health and safety policies OK3. Organization environmental policies OK4. Organization regulations and global best practices

B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Animal Breeding equipment and tools, including their designs, uses, repair, and maintenance. - TK2. Animal Breeding data mining analysis software - TK3. Animal organisms, their tissues, cells, functions, species, breeds, interdependencies, and interactions with each other and the environment. - TK4. Principles and methods of promoting and selling products or services. This includes marketing strategy and tactics, product demonstration and sales techniques - TK5. Administrative and clerical procedures and systems such as word processing, managing files and records - TK6. Entering, transcribing, recording, storing, or maintaining information in written or electronic form.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Government regulatory agency requirements for Animal welfare, health & safety.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: - CS1. write in English and give simple concise instructions. - CS2. Record data of all observed traits of interest - CS3. Record all activities undertaken and challenges faced - CS4. Write and manage a personal daily task list as well as for subordinates
	Reading Skills
	The individual on the job must be able to: - CS5. Read and interpret internal/external documents. - CS6. Read and understand manuals, health and safety instructions, memos, other company documents. - CS7. Read from different sources- books, journal publications, research articles on Animal Breeding.
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : - CS8. Express statements or information clearly so that others can hear and understand. - CS9. Communicate clearly during stakeholder round table meetings - CS10. Participate in and understand the main points of simple discussions regarding Animal Breeding. - CS11. Respond appropriately to any queries regarding Animal Breeding.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process.

	PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan for procurement of livestock for breeding in a timely manner PS4. Plan to arrange for necessary materials to be in place PS5. Plan and organise work to meet deadlines. PS6. Plan to have up-to-date records at all times PS7. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS8. Follow code of conduct. PS9. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS10 Recognize problems and search for solutions. PS11. Choose best methods to complete assigned tasks. PS12. Approach relevant authority when required. PS13. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS14. Apply domain knowledge, observations and data to select course of action to perform tasks PS15. Monitor breeding conditions and breeding livestock and suggest improvements if any.
	Critical Thinking
	The individual on the job must be able to: PS16. Use common sense and make judgements on day to day basis PS17. Critically evaluate information obtained from stakeholders and co-workers to perform day to day activities. PS18. Use reasoning skills to identify and resolve basic problems

UNIT 4 [This unit is about Safety, Hygiene and Environmental Sanitation Practices for Breeding Operations]

Unit No.	04
Unit Title	Safety, Hygiene and Environmental Sanitation Practices for Breeding Operations
Description	This unit is about maintaining safety, health and environmental protection for the individual, the animals and the environment
Scope	This unit covers the following: • Health & safety regulations • Environmental protection regulations • Animal welfare regulations
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health & Safety Regulations	To be competent, the individual must be able to: - PC1. Supervise implementation of national and organizational safety and health policies and regulations. - PC 2. Assess risks and possible safety hazards of all aspects of operations - PC3. Ensure all standard scientific guidelines related to medicine administration procedures are followed e.g. use of separate syringes for different vaccines - PC4. Ensure only permissible products (that are health friendly) are administered
Environmental Protection Regulations	To be competent, the individual must be able to: - PC5. Supervise implementation of environmental standard operating procedures and policies of the organisation - PC6. Demonstrate implementation of national and global environmental regulations. - PC7. Delegate appropriate measures for carcass and waste disposal - PC8. Ensure only permissible products (that are environmentally friendly) are administered
Animal Welfare Regulations	To be competent, the individual must be able to: - PC9. Ensure all scientific guidelines related to medicine administration procedures are followed e.g. use of separate syringes for different vaccines - PC10. Ensure non-cruelty practices towards animals - PC11. Demonstrate implementation of national and global animal welfare regulations
Knowledge and Understanding (K)	

A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Organization Quality, health and safety policies OK2. Organization environmental policies OK3. Organization regulations and global best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Safety and health risk assessment TK2. Environmental Risk assessment TK3. Toolbox talk
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Government regulatory agency requirements for health & safety
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, health and safety instructions, memos, other company documents. CS4. Read from different sources- books, screens in machines and signage. CS5. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
B. Professional Skills	The individual on the job must be able to : CS6. Express statements or information clearly so that others can hear and understand. CS7. Participate in and understand the main points of simple discussions. CS8. Respond appropriately to any queries.
	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity

	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7 Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to the following; Animal identification tags, pregnancy diagnosis kits, OB lube for rectal lubrication, OB sleeves, mounting detectors, disinfecting solutions, scrotal tape measure, paint sticks, CIDRs, computer, printer and relevant software programs, personal protective equipment (PPE), weighing scales, record book, feed, water facilities, crush pens, company's standard operating procedures, reporting template, calculators, disposable gloves, collection tubes, plastic tubes (for storage of samples), adhesive labels and markers.

6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER

Dilemmas associated with the job of an Animal Breeder - Geneticist include resistance by farmers to adopt new techniques, handling large populations of animals at a go, long working hours, driving long distances, poor road infrastructure, exposure to dangerous chemicals, time pressure to complete tasks with desired output, working in extreme weather such as hot and cold conditions, working in noisy, wet and dusty environments, language barriers, unrealistic expectations from farmers and stakeholders, insufficient funding, outbreak of diseases, breeds of stunted growth.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to dilemmas include ensuring good time management and planning so as to manage large animal populations, Updating records frequently and efficiently managing task list every day, consulting extensively where unclear, paying attention to detail, adhering to company's safety and standard operating procedures at all times, consulting on safety guidelines where unclear, participating in workplace safety sensitization and awareness, wearing protective clothing, exercising regularly to maintain physical fitness, managing work stress etc

7. WORKING CONDITIONS/ENVIRONMENT

Working conditions include working in remote areas, stand/walk for long hours, working at odd hours, working in laboratory environment, working in cold, dusty, hot and wet conditions, dealing with donors or funders, working with Animal Breeding software and data manipulation software, in lab environments.

8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

8.1 Internal/Within the Organization

Parties involved/interacting with the job holder who are internal to the organization include supervisors, technicians, trainers and other employees.

8.2 External/Outside the Organization

Parties involved/interacting with the job holder who are external include customers/clients, Farmers, government regulators, trainers, suppliers of equipment/tools/consumables, Animal Breeders from other organizations, donor agencies, non-governmental organizations, occupational health and safety associations, Academia etc.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to sustain strenuous conditions;
- Be able to walk and stand for long periods of time;
- Bend, stretch, twist, or reach out;
- Be able to manage/lift relatively heavy animals materials, tools and equipment;
- Be able to use fingers, hands and feet with ease to complete the assigned task (dexterity);
- Be able to withstand feelings of wetness, itching, or coldness on the body for
- Long periods of time;
- Sound mental health.

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programmes developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programmes developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.AB.01		
ZQF Level	9	Version Number	01
Sector	Agriculture	Date of Approval	19 th May 2022
Sub Sector	Livestock (Animal Breeding)	Date of Last Review	N/A
Occupation	Animal Breeding	Date of Next Review	May 2027

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